

ProMix DJ Studio

The Ultimate Browser-Based Party Mixer

USER MANUAL · Version 1.0

Dual Decks ● BPM & Key Detection ● Camelot Wheel ● YouTube Search ● MP3 Download ● Mix Recording ●
3-Band EQ ● 5 Effects ● Auto Mix ● Stem Separation

Table of Contents

1. Overview & Feature List
2. System Requirements
3. Quick Start Guide
4. Interface Layout
5. Loading Tracks
6. Playback Controls, Hot Cues & Loops
7. BPM & Key Detection
8. Camelot Wheel — Harmonic Mixing
9. EQ & Mixing
10. Crossfader
11. Effects
12. YouTube Search & MP3 Download
13. Recording & Exporting Your Mix
14. Auto Mix
15. Stem Separation
16. Sync & Pitch
17. Keyboard Shortcuts
18. Tutorial 1 — Your First Mix
19. Tutorial 2 — DJ Party Set with Recording
20. Tutorial 3 — YouTube to Mix
21. Tutorial 4 — Harmonic Mixing
22. FAQ
23. Troubleshooting

1. Overview

ProMix DJ Studio is a fully self-contained DJ mixing application delivered as a single HTML file. Open it in any modern web browser — no installation, no app store, no subscription. Everything runs locally using the Web Audio API.

It was designed by analyzing DJ.Studio, Algoriddim djay Pro, VirtualDJ, and Fadr — combining the best features of all four into one free, portable tool.

Feature Summary

Feature	Description
Dual Decks	Two independent decks (A and B) with animated vinyl turntables
Waveform Display	Real-time waveform with click-to-seek and playhead position
BPM Detection	Automatic beats-per-minute analysis on every loaded track
Key Detection	Musical key detection mapped to Camelot Wheel notation
Camelot Wheel	Interactive harmonic mixing guide for compatible key transitions
3-Band EQ	Hi / Mid / Lo equalizer per deck with rotary knobs
Crossfader	Linear, Scratch, and Smooth curve modes
5 Effects	Reverb, Delay, Filter, Flanger, Echo — all with parameter knobs
Hot Cues	8 hot cue points per deck — set and jump with one click
Loop Control	4-beat auto-loop with one button
Pitch / Sync	±8% pitch slider, key shift (±semitones), one-click BPM sync
Auto Mix	One-click BPM-synced crossfader transition
YouTube Search	Search by song/artist, load results directly to a deck
MP3 Download	Download any YouTube track as MP3 via cobalt.tools
Mix Recording	Record the master output and download as an audio file
Stems Panel	Vocal/Drum/Bass/Melody controls + AI stem separation via Fadr

2. System Requirements

Item	Requirement
Browser	Chrome 90+, Firefox 88+, Edge 90+, Safari 15+
OS	Windows 10/11, macOS 11+, Linux
RAM	4 GB min, 8 GB recommended
Audio Files	MP3, WAV, FLAC, OGG, AAC, M4A
Internet	Only for YouTube search and MP3 download

Item	Requirement
API Key	Optional — enables YouTube in-app search (free)

- **Best Experience:** Use Google Chrome for the best Web Audio API performance.

3. Quick Start Guide

Get mixing in under 2 minutes:

- 1 Open the file**
 Download dj-mixer.html from GitHub. Double-click to open in your browser.
- 2 Load Track A**
 Click ■ Load on Deck A. Select any MP3 or audio file. Waveform appears and BPM/key are auto-detected.
- 3 Load Track B**
 Click ■ Load on Deck B. Load a second track.
- 4 Play Deck A**
 Press the green ■ Play button or press Space on your keyboard.
- 5 Sync and mix**
 Press SYNC on Deck B to match tempo. Press Play on Deck B. Slide the crossfader to blend.
- 6 Record your mix**
 Click ● REC before mixing. Click ■ Download Mix to save when done.

- **You're mixing!** That's all it takes. Read on to explore all professional features.

4. Interface Layout

The interface is organized in a professional DJ layout. Below is a schematic overview:

[illegible]

Deck Controls Reference

Control	Function
■ Load	Open file browser to load an audio file
Vinyl Canvas	Animated spinning record
Waveform	Click anywhere to jump to that position
BPM	Auto-detected tempo (use TAP to correct)
KEY	Camelot notation key (e.g. 8A = A minor)
■ Play	Start or pause playback
CUE	Set cue point (paused) or jump to it (playing)
SYNC	Match BPM to the other deck
LOOP	Activate a 4-beat loop from current position
Hot Cues 1-8	Set and jump to up to 8 saved positions
PITCH slider	Adjust playback speed $\pm 8\%$
Key Shift –/+	Shift pitch by semitones

5. Loading Tracks

- Click ■ Load**
The file browser opens. Navigate to your music folder.
- Select your audio file**
Supported: MP3, WAV, FLAC, OGG, AAC, M4A.
- Wait for analysis**
The waveform appears and BPM/key are detected. Takes 1–3 seconds.
- Check the readout**
If BPM looks wrong, use TAP tempo to correct it.

- **Large files:** Files over 100 MB may take several seconds. MP3s at 320kbps load fastest.
-

6. Playback Controls, Hot Cues & Loops

Action	How To
Play / Pause	Click ■ or press Space (Deck A) / Enter (Deck B)
Seek forward 5s	Press → (Deck A)
Seek back 5s	Press ← (Deck A)
Jump to position	Click anywhere on waveform or timeline
Set Cue Point	Pause at desired position, click CUE
Jump to Cue	Click CUE while playing

Hot Cues

- 1 Navigate to the spot to mark**
For example, the start of the chorus.
- 2 Click an empty Hot Cue button (1–8)**
The button changes color. Position is saved.
- 3 Click the colored button to jump**
Instantly jumps to that saved position at any time.

- **Pro Tip:** Set Hot Cue 1 at intro, Cue 2 at first drop, Cue 3 at breakdown for fast navigation.

Loop Controls

Action	How To
Start loop	Click LOOP during playback — activates 4-beat loop from current beat
Exit loop	Click LOOP again — returns to normal playback

7. BPM & Key Detection

ProMix analyzes audio using onset energy detection to determine tempo. BPM is clamped to 70–200 BPM to handle half-time and double-time tracks correctly.

Correcting BPM with Tap Tempo

- 1

Play the track
Listen to the beat.
- 2

Click TAP on the beat
Click 4+ times. BPM updates after each tap.
- 3

Tap after 2+ seconds to reset
A 2-second gap clears previous taps.

Key Detection

ProMix uses a pitch class profile (Krumhansl–Schmuckler model) to detect key. Result is shown in Camelot Wheel notation (e.g., 8A = A minor, 8B = C major).

- **Camelot Notation:** Numbers 1–12 represent key families. A = minor, B = major. Adjacent numbers are compatible.

8. Camelot Wheel

Open the Camelot Wheel from the header to see all 24 musical keys arranged by harmonic compatibility.

Position	Meaning
Same number (e.g., 8A+8B)	Relative major/minor — very compatible
Adjacent number (e.g., 8A+7A)	Neighboring key — smooth and natural
±2 positions	Slightly more tension but can work
Opposite side of wheel	Clashing keys — avoid for smooth mixes
Outer ring (B keys)	Major keys
Inner ring (A keys)	Minor keys

- 1

Note the key of your current track
Example: Deck A shows 8A (A minor).
- 2

Open the Camelot Wheel
Find position 8A.
- 3

Choose a compatible key for Deck B
Compatible: 8B (C major), 7A (D minor), 9A (E minor).

4

Mix in harmonically

The transition sounds musically natural.

9. EQ & Mixing

Band	Frequency	Controls
HI (High shelf)	10,000 Hz+	Cymbals, hi-hats, air
MID (Peaking)	~1,000 Hz	Vocals, guitars, presence
LO (Low shelf)	200 Hz and below	Kick drum, bass, sub

Drag knobs up to boost (+12 dB max), down to cut (−12 dB max). Scroll the mouse wheel for precise control. Center = flat.

- **Classic EQ Technique:** Cut the LO on Deck B before mixing in. As you crossfade to B, gradually bring the lows back up. Prevents bass clash.

10. Crossfader

Far left = only Deck A. Far right = only Deck B. Center = equal blend.

Mode	Best For	Behavior
Linear	General mixing	Smooth, proportional blend
Scratch	Hip-hop, turntablism	Sharp cut near edges
Smooth	Deep house, ambient	Cosine curve, very gradual fade

11. Effects

- **FX Deck Routing:** Use the A/B buttons in the FX panel to select which deck receives the effects.
-

REVERB

Adds room/hall ambience.

Knob	Function
Wet	How much reverb is mixed in
Size	Size of the virtual room

DELAY

Creates repeating echoes.

Knob	Function
Time	Delay time 0–1.5 sec
Feedback	How many repeats (0=once, max=infinite)

FILTER

Bandpass filter — great for buildups.

Knob	Function
Freq	Center frequency 80 Hz–16 kHz
Q	Filter width (higher = narrower)

FLANGER

Swirling modulation effect.

Knob	Function
Rate	Speed of oscillation
Depth	Intensity

ECHO

Single-repeat accent echo.

Knob	Function
Beat Div	Echo timing relative to beat

12. YouTube Search & MP3 Download

Getting a Free YouTube API Key

1 Go to Google Cloud Console

Visit console.cloud.google.com and sign in with your Google account.

2 Create a new project

Click project dropdown → New Project → name it "DJ Mixer" → Create.

3 Enable YouTube Data API v3

APIs & Services → Library → search "YouTube Data API v3" → Enable.

4 Create credentials

APIs & Services → Credentials → Create Credentials → API Key. Copy the key.

5 Paste it in ProMix

Paste into the API Key field in the header. Searches are now live.

- **Free quota:** 10,000 search units per day — roughly 100 searches. No credit card required.
-

Downloading MP3s

1 Search for a track

Type in the search bar and press Enter.

2 Click ■ MP3

The app requests a download from cobalt.tools (free, no account).

3 The file downloads

Saved to your Downloads folder.

4 Load into ProMix

Use ■ Load to load your downloaded MP3.

- **Important:** Only download music you have the right to use. Respect copyright law.
-

13. Recording Your Mix

- 1 **Click ● REC before mixing**
Button turns red and pulses. Recording starts.
- 2 **Perform your mix**
Everything you do is captured.
- 3 **Click ■ STOP REC when finished**
Mix is stored in memory.
- 4 **Click ■ Download Mix**
File downloads as ProMix_[timestamp].webm.

Converting to MP3

- VLC Media Player — Media → Convert/Save → Add file → choose MP3
- Audacity — File → Import → export as MP3
- Online: cloudconvert.com (free)
- FFmpeg: `ffmpeg -i ProMix_123.webm output.mp3`

14. Auto Mix

■ Auto Mix automates the transition: syncs BPM and smoothly moves the crossfader over ~4 seconds.

- 1 **Load tracks on both decks**
Make sure both tracks are loaded and BPM is detected.
- 2 **Start playing Deck A**
Let it reach the point where you want to mix out.
- 3 **Click ■ Auto Mix**
App syncs Deck B to Deck A's BPM, starts Deck B, then fades the crossfader.

15. Stem Separation

Stems are the individual components of a song: Vocals, Drums, Bass, and Melody. Separate them to create acapellas, instrumentals, or mashups.

- 1 **Click ■ Separate Stems (AI)**
Opens Fadr.com in a new tab.
- 2 **Upload your track to Fadr**
Drag and drop your audio file.

3 Wait for AI processing

Under 60 seconds. Splits into up to 16 stems.

4 Download stems

Download acapella, instrumental, or any individual stem.

5 Load stems into ProMix

Load acapella on Deck A + instrumental on Deck B = instant mashup!

16. Sync & Pitch

Control	Function
SYNC Button	Matches this deck's BPM to the other deck
Pitch Slider	Adjusts speed -8% to +8%
Key Shift -/+	Shifts pitch one semitone per click

17. Keyboard Shortcuts

Shortcuts do not fire when cursor is in a text input.

Key	Action
Space	Play / Pause Deck A
Enter	Play / Pause Deck B
←	Seek Deck A back 5 seconds
→	Seek Deck A forward 5 seconds
Q	Set/Jump Cue on Deck A
W	Set/Jump Cue on Deck B
S	Sync Deck B to Deck A's BPM
A	Toggle Mix Recording

18. Tutorial 1 — Your First Mix

Goal: Load two tracks and perform a manual crossfader mix. Time: ~5 minutes.

- 1 Open dj-mixer.html in Chrome**
Double-click the file. Loads instantly.
- 2 Load a track on Deck A**
Click  Load. Choose an upbeat song with a steady kick drum.
- 3 Load a different track on Deck B**
Choose something with similar energy and tempo.
- 4 Move crossfader all the way left (A)**
Deck A is the only thing heard.
- 5 Press Space to play Deck A**
Listen and enjoy.
- 6 Click SYNC on Deck B**
Matches Deck B's tempo to Deck A.
- 7 Position Deck B at the start**
Click far left of Deck B's waveform.
- 8 Start Deck B at a phrase boundary**
When Deck A reaches a chorus, press Enter.
- 9 Slowly slide crossfader to the right**
Over 8–16 beats, fade from A to B. First mix done!
- 10 Pause Deck A**
Load your next track and repeat.

19. Tutorial 2 — Party Set with Recording

Goal: Record a 3-track mix with effects. Time: ~15 minutes.

- 1 Prepare your tracklist**
Choose 3 songs ~120–130 BPM. Check Camelot keys for compatibility.
- 2 Click  REC to start recording**
Button pulses red.
- 3 Load Track 1 on Deck A and play it**
Press Space. Let it run for a minute.
- 4 Load Track 2 on Deck B**
Sync it. Position at intro or drop.

5 Apply a build-up effect
Enable FILTER on Deck A. Sweep Freq knob upward.

6 Cut the bass on Deck A (LO knob)
Drag Deck A's LO knob counter-clockwise.

7 Start Deck B and move crossfader
Slowly push crossfader right over 8 bars.

8 Load Track 3 on Deck A and repeat
Deck A is free. Repeat the transition process.

9 STOP REC and Download Mix
Your 3-track mix is saved.

20. Tutorial 3 — YouTube to Mix

Goal: Search YouTube, download two MP3s, and mix them. Requires API key.

- 1 Enter your YouTube API key**
Paste into the header API key field.
- 2 Search for your first song**
E.g., "Calvin Harris Summer". Press Enter.
- 3 Click ■ MP3 and save**
Download the file to your computer.
- 4 Search for your second song**
E.g., "Avicii Wake Me Up". Download as MP3.
- 5 Load both MP3s into the decks**
■ Load → Track 1 to Deck A, Track 2 to Deck B.
- 6 Check harmonic compatibility**
Open the Camelot Wheel. Use Key Shift if needed.
- 7 Record and download your mix**
Click ● REC, mix, then ■ Download Mix.

21. Tutorial 4 — Harmonic Mixing

Goal: Build a harmonically perfect set using the Camelot Wheel.

- 1 Load your first track on Deck A**
Note the KEY display. Example: 8B (C major).
- 2 Open the Camelot Wheel**
Find 8B. Compatible: 8A (A minor), 7B (F major), 9B (G major).
- 3 Find a track in a compatible key**
Load candidates on Deck B, check detected key.
- 4 Use Key Shift if close but not exact**
If Deck B shows 9A instead of 8A, click – once.
- 5 Mix in Deck B**
The transition sounds musically natural.
- 6 Plan ahead around the wheel**
From 8A, next options are 8B, 7A, 9A — take the audience on a journey.

22. FAQ

No sound is coming out

Browsers require a user click before playing audio. Click anywhere on the page first. Check system volume and that the browser tab is not muted.

BPM is half or double the actual tempo

Use the TAP button to correct it manually — tap 4–8 times on the beat.

YouTube search returns an error

API key may be invalid or over quota (10,000 units/day free). Check Google Cloud Console.

MP3 download does not start

cobalt.tools may be temporarily busy. Try again, or click  YT and use another download tool.

Can I use this offline?

Yes — all core mixing features work offline. Internet only needed for YouTube search/download.

Can I save hot cues between sessions?

Not currently. Hot cues and settings reset when you refresh the page.

The file does not open on my computer

Right-click dj-mixer.html → Open with → choose Chrome, Firefox, or Edge.

23. Troubleshooting

Issue	Solution
App is silent	Click anywhere on page. Check volume. Unmute browser tab.
Waveform does not appear	Wait 2–3 sec. Try converting to MP3 if it persists.
BPM says "--"	Use TAP tempo to set it manually.
Key says "--"	Click the middle of the waveform then reload.
Effects have no effect	Check FX toggle is active (purple) AND correct deck is selected.
YouTube search is blank	Verify API key. Try disabling VPN or ad-blocker.
Download Mix is tiny	You must click  REC before starting your mix.
Tracks out of sync	Correct BPM with TAP, then click SYNC again.
App slow on long tracks	Use MP3 format. Very long WAV files may be slow.